

DOC-02

Entity Relationship Diagram

Database schema with all 17 tables and relationships



Project	GPA-ERP V5.0
Type	Construction Cost Control ERP
Stack	FastAPI · Next.js 14 · PostgreSQL
Version	V5.0

ERD (Mermaid)



Table	Rows (expected)	Key Indices
roles	7 (fixed enum)	name UK
users	Low (10–100)	email UK, role_id
app_menus	~12 (fixed)	key UK, section+sort_order
user_menu_permissions	users × menus	user_id+menu_id UK
projects	Medium (10–500)	code UK, status
cost_codes	Low (20–200)	code UK, self-ref parent_id

Table	Rows (expected)	Key Indices
cost_centres	Low (5–50)	code UK
approval_rules	Very low (4–20)	priority, is_active
project_documents	Medium	project_id+doc_type
account_receivables	Medium (100–10k)	project_id+status, invoice_no
expenses	High (1k–100k)	project_id+status, current_approver_role, submitted_by
petty_cash_reports	Medium	project_id+month, status
petty_cash_report_lines	High	report_id+line_no
legal_documents	Low-medium	doc_number UK, status, doc_type
inventory_items	Low (10–1000)	code UK, category
inventory_txns	High	item_id, project_id
audit_logs	Very high (append-only)	entity_type+entity_id, changed_by, created_at